

B.Tech. Degree VI Semester Special Supplementary Examination January 2017

CS/IT 1602 COMPILER CONSTRUCTION (2012 scheme)

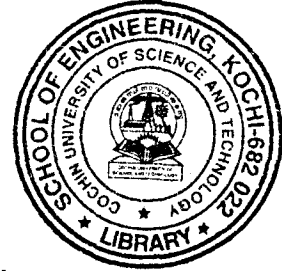
Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Explain the role of regular definition in the specification of tokens.
- (b) Write a short note on lexical analyzer tool 'LEX'.
- (c) Differentiate sentence and sentential form of a grammar.
- (d) Discuss any two error recovery strategies that a parser can use.
- (e) Differentiate synthesized and inherited attributes.
- (f) Explain the use of dependency graph in the translation of an input string.
- (g) Discuss the different structure preserving transformations on a basic block.
- (h) Explain any two code optimization techniques with examples.



PART B

(4 × 15 = 60)

- II. Discuss the phases of a compiler with a neat diagram. Show the representation of the following statement in each phase of a compiler. (15)
Position: = initial + rate * 60.

OR

- III. What is the role of transition diagram in the construction of a lexical analyzer? (15)
Draw the transition diagram for:
(i) Relational operators (ii) unsigned numbers.

- IV. (a) What is left recursion? Discuss the algorithm to eliminate left recursion from a grammar. Give example. (8)
- (b) Show that the following grammar is not LL(1): (7)
 $S \rightarrow iEtSS' | a$
 $S' \rightarrow eS | \epsilon$
 $E \rightarrow b$

OR

(P.T.O.)

- V. (a) What is an operator grammar? Give example. (5)
(b) Show that the following grammar is not SLR(1): (10)
 $S \rightarrow L = R$
 $S \rightarrow R$
 $L \rightarrow *R$
 $L \rightarrow id$
 $R \rightarrow L$
- VI. What is type checking? Explain the type checking of expressions, statements and functions. (15)
- OR**
- VII. Discuss the different storage allocation strategies. (15)
- VIII. Discuss the different intermediate representations. Explain the implementations of three address statements. (15)
- OR**
- IX. Explain the issues in the design of a code generator. (15)